

Virtual Reality with Moodle for Training in Telemedicine

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Abstract. *Background:* Telemedicine is increasingly integrated into healthcare, requiring robust training for healthcare professionals. Traditional training methods often lack realism and immersion, highlighting the need for innovative approaches. *Objectives:* This study aims to integrate Virtual Reality with Moodle to enhance telemedicine education, focusing on chronic cardiac conditions. *Methods:* A VR-based training system was developed using Unity and WebXR, integrated with Moodle. The system includes a serious game with virtual patients created using Synthesia AI. Telemedicine experts configure training sessions via Moodle, while learners access preparatory materials and engage with the VR environment. *Results:* A usability test involving 12 participants showed high satisfaction scores (94.42% for learners, 98.29% for experts). *Conclusion:* This study demonstrates the potential of integrating VR with Moodle to innovate telemedicine education. The approach offers personalized and accessible training experiences for healthcare professionals to improve telemedicine skills.

Keywords. Telehealth, Educational Virtual Reality, Medical Field Training, Learning Health System.

1. Introduction

Telemedicine has deeply changed healthcare delivery by leveraging digital technologies to provide remote medical services. Telemedicine is increasingly integrated into clinical practice, aiming to improve healthcare accessibility, enhance treatment adherence, and reduce travel requirements optimizing cost-effectiveness [1][2]. Italy, through incentives and investments, is promoting e-health, opening up challenges in the training of health personnel. As telemedicine is becoming a permanent component of healthcare processes, it is essential to provide robust training to enhance skill acquisition among healthcare personnel [3]. To meet the growing demand of remote care, healthcare professionals must acquire both technical skills (e.g., platform navigation, remote diagnostics) and non-technical skills (e.g., virtual communication and patient engagement). However, there is a lack of published evidence on effective delivery of virtual consultation training despite its increasing use [4].

Traditional simulation-based training methods have long been used for experiential learning in healthcare. These approaches are based on standardized patients, who are trained individuals simulating specific clinical scenarios [5]. In the telemedicine context,

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such approaches have been adapted to target specific digital competencies. For instance, Mulcare et al. [6] developed a training course comprising an introductory phase, simulations and group discussions to test history-taking, virtual physical examinations, and patient disposition planning. Similarly, Nash et al. [7] introduced a telemedicine communication program emphasizing equity and patient access, demonstrating improvements in targeted skills and physician satisfaction. However, while these traditional approaches effectively address specific skills, they often lack the realism and immersion provided by technologies [8].

The latest trend in medical simulation uses serious games (SGs) with virtual patients through desktop platforms, Virtual Reality (VR), or Augmented Reality (AR), enabling learners to train in diverse simulated clinical environments for real-world scenarios [9]. VR simulations enable medical students and healthcare professionals to practice upcoming procedures in a controlled, virtual environment [10][11]. In contrast to traditional training, VR SGs improve personalized learning, real-time performance evaluation, and greater accessibility for training in remote areas, overcoming the limitations of traditional methods. Learning Management Systems (LMSs) can address and overcome limitations in VR training delivery by facilitating course management and training via web applications [12]. Widely adopted in universities and institutions, LMSs enable instructors to manage content, design exercises, and administer assessments, while students access multimedia materials and complete exams using web-enabled devices. Popular platforms include Moodle, Blackboard, and Canvas, with Moodle [13] standing out for its open-source accessibility. Moodle is one of the most popular choices for many medical schools and institutions, to enhance teaching methodologies [14]. However, despite the growing need for improved healthcare workforce training, both VR and LMSs remain underutilized in the telemedicine sector [15], highlighting a gap that must be addressed to maximize their potential.

This research study presents an innovative approach that integrates immersive VR with Moodle system to create an accessible, dynamic telemedicine training experience. Leveraging recent advances in VR, the developed solution aims to simulate clinical scenarios for medical personnel to practice telemedicine skills. The combination of these technologies aims to define a personalized learning experience, enhance training effectiveness and provide a safe, engaging environment to practice telemedicine.

2. Methods

This section outlines the Moodle-integrated training workflow and describes its technological framework, including Unity, WebXR, and a custom PHP module.

2.1. Workflow to use the e-learning course on Moodle

The designed Moodle course provides preparatory materials for conducting telemedicine with patients suffering from chronic cardiac conditions, such as arterial hypertension and hyperlipidemia, enabling users to take on specific roles for managing and utilizing the content of an e-learning course. Two user profiles get access to the telemedicine training course: teachers and students.

Teachers, who are experts in telemedicine, curate learning materials (e.g., clinical case studies and guidance for using the VR SG and configure the parameters of VR training sessions, which are categorized into three difficulty levels: “1 - Beginner,” “2 -

Intermediate,” and “3 - Advanced” via a dedicated Moodle interface (Figure 1a). After completing the setup, they can monitor learners' progress in using the VR SG and adjust the training parameters accordingly (Figure 1b).

Students are medical learners who enroll in the course to improve their skills in conducting televisits with cardiac patients. After acquiring sufficient theoretical knowledge using the Moodle course, they access the VR SG through a web link within the Moodle course, which redirects them to the online VR simulation. The virtual environment replicates a medical office where learners use natural gestures to interact with a virtual computer and consult with digital patients.

The SG provides an immersive telemedicine training experience, guiding medical learners through the complete televisit process. The scenario begins in a detailed virtual office, where students review patient records and vital signs before initiating a remote consultation. The televisit unfolds through a 2D interactive interface embedded within the 3D office environment, simulating a physician's desktop. Users interact with virtual patients by selecting appropriate responses, activating necessary tools (e.g., video and audio communication), and ensuring compliance with telemedicine protocols such as privacy confirmation and report verification. The game dynamically pauses at key moments, prompting learners to make clinical decisions and reinforcing critical thinking.

To enhance realism, five virtual patients have been designed meticulously, each with a detailed clinical profile, including medical history, current treatments, and symptom descriptions. The patient cohort consists of three females and two males aged 45-70 years, with physician-patient dialogues carefully crafted to simulate real teleconsultations. Performance assessment is a key aspect of the game: players start with a score of 100 points, with penalties applied for errors such as failing to activate essential functions, selecting incorrect responses, or missing procedural steps. The severity of penalties increases with difficulty level, and repeated mistakes lead to progressively higher deductions. At the end of each session, learners receive a performance-based rating ranging from “Very Low” (<40 points) to “Excellent” (90-100 points), providing immediate feedback on their proficiency.

All performance data, including execution time and procedural correctness, are tracked and recorded during each televisit with a randomly assigned virtual patient (Figure 2). Upon completing the VR session, this data is automatically transmitted to the Moodle server to be available to the telemedicine experts.

2.2. Technological framework to embed the VR serious game in Moodle

Three technological features have been developed specifically to integrate in Moodle the VR SG with the functionalities described in §2.1:

- **VR SG via Web Application.** The VR SG is developed with Unity and seamlessly integrated into Moodle using WebXR. WebXR is a web standard enhancing accessibility by enabling immersive VR experiences directly through web browsers using a VR Head Mounted Display (e.g., Meta Oculus Quest and HTC Vive).
- **PHP Module for Data Exchange.** A dedicated PHP module has been developed to retrieve the SG parameters at the end of each VR training session, ensure secure authentication, and record user performance, including scores and difficulty levels, for teacher's assessment. Additionally, teachers can use the

module to personalize training by choosing the level of difficulty of VR sessions for each learner.

- Virtual Patients via Generative AI. Virtual patients are created using Synthesia, an AI platform that generates customizable videos with multilingual speech and facial expressions. These videos are integrated into the VR SG to enhance realism in teleconsultations. A C# component in Unity dynamically retrieves and displays randomly selected virtual patients.

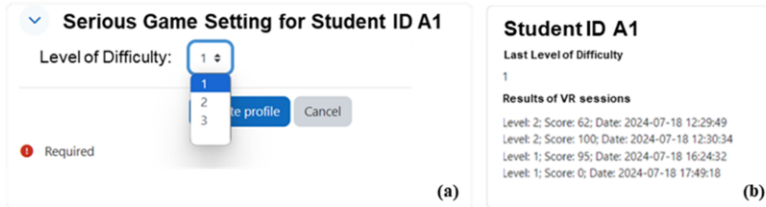


Figure 1. Parts of the custom Moodle interface developed for (a) serious game settings and (b) monitoring of medical learners. The ad-hoc developed Moodle interface is accessible exclusively to telemedicine experts.

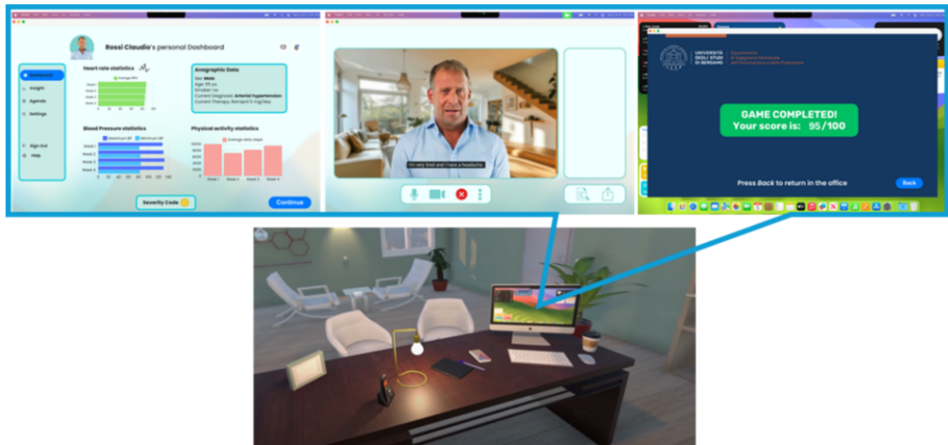


Figure 2. Telehealth consultation in the VR medical office to interact with patient dashboards, monitor health metrics, and conduct realistic patient interviews.

3. Results

A usability test was conducted to evaluate the effectiveness and user experience of the proposed VR-based telemedicine training system integrated with Moodle. The study involved 12 participants with diverse occupational backgrounds to capture comprehensive feedback on system usability. The participant pool consisted of 2 physicians, 3 nurses, 2 medical students, and 5 external users without specific healthcare experience. Notably, none of the participants had prior exposure to VR systems, ensuring unbiased feedback focused on usability aspects.

- Learners Group (9 testers: a physician, 2 nurses, 2 medical students and 4 external users). This group engaged directly with the SG and explored the Moodle course content, including theoretical materials, quizzes, and feedback sections.

- Experts Group (3 testers: a physician, a nurse, and an external user). This group is responsible of teaching tasks in Moodle, including managing materials, posting announcements, setting difficulty levels of the SG, and reviewing student scores.

To assess usability, participants completed the Post-Study System Usability Questionnaire (PSSUQ) after their tasks. The PSSUQ was chosen for its reliability in evaluating system usability [16] using a set of questions subdivided into three categories: System Usefulness (SysQual, Q1-Q6), Information Quality (InfoQual, Q7-Q12), and Interface Quality (IntQual, Q13-Q15). A 7-point Likert scale was used, with 7 indicating the highest level of agreement.

4. Discussion

The collected quantitative results demonstrated high user satisfaction. Average scores for each category were calculated, and the overall satisfaction scores were 93.6% for learners and 97.6% for experts. These values significantly exceeded the predefined satisfaction threshold of 80%, indicating strong usability across both groups. Figure 3 illustrates the average statistics for the learners (green) and experts (blue) compared to the satisfaction threshold. The high scores confirm that the system was well-received by both learners and experts, emphasizing its effectiveness in delivering telemedicine training.

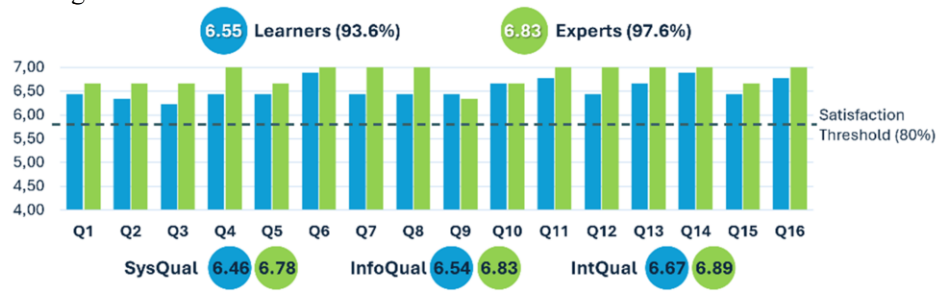


Figure 3. Average Statistics – Learners (green) and Experts (blue) - with respect to the satisfaction threshold.

Beyond numerical scores, qualitative feedback gathered from the questionnaire’s comment section highlighted specific areas requiring improvement. One of the most frequently mentioned issues was related to the VR controller buttons during virtual televisit sessions. Learners noted that the visual feedback from button presses was delayed, which negatively impacted the interaction flow. They expected an immediate response, and the lack of real-time feedback was perceived as suboptimal. This observation aligns with existing literature on VR usability, where real-time feedback is a critical component of effective interaction design [17]. Addressing this issue by optimizing the controller response mechanism could enhance user experience and increase engagement.

5. Conclusion

This study highlights the effectiveness of integrating VR with Moodle for telemedicine training. The serious game leverages immersive VR features and structured learning to

create a personalized and accessible training experience. Usability testing revealed high satisfaction scores, confirming its value for medical learners and experts. While minor refinements are needed to improve interaction mechanics, the system establishes a strong foundation for future advancements in VR-based telemedicine education. Future research should focus on refining user interactions and expanding virtual patient scenarios to enhance realism and efficacy.

Fundings: This work is partially funded by the National Plan for NRRP Complementary Investments - project n. PNC0000003 - AdvaNced Technologies for Human-centrEd Medicine (ANTHEM).

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